

## Exercise 9

1. What is the Mean time to failure values of different RAID systems?

- a. RAID 0 with 2 disks
- b. RAID 2 with 2 disks
- c. RAID 1 with 2 disks
- d. RAID 1 with 3 disks
- e. RAID 3 with 3 disks
- f. RAID 4 with 3 disks
- g. RAID 5 with 3 disks
- h. RAID 6 with 5 disks

2. Which of the following RAID configurations that we saw in class has the lowest disk space utilization? Your answer needs to have explanations with calculations for each case.

- (1) RAID 0 with 2 disks
- (2) RAID 1 with 2 disks
- (3) RAID 3 with 3 disks

Where does this lack of utilization of space go, i.e., where we can use such a configuration as it has some benefits gained due to the loss of space utilization?

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

```
10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15
```